

Inverse Trigonometry — Problem Set

Warm-up

1. Evaluate $\sin^{-1}\left(\frac{1}{2}\right)$.
2. State the domain and range of the function $y = \cos^{-1} x$.
3. Find the domain of $y = \cos^{-1}(3x)$.
4. Evaluate $\tan^{-1}(-1)$.
5. For what values of x is $y = \sin^{-1}(x + 5)$ defined?
(i.e What is the domain of the function?)
6. State the range of $y = 3 \sin^{-1} x$.
7. Solve for x : $\sin^{-1} x = \frac{\pi}{6}$.
8. Find the domain of $y = \tan^{-1}(2x - 4)$.
9. Evaluate $\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right)$.
10. What is the horizontal asymptote of $y = \tan^{-1} x$ as $x \rightarrow \infty$?

Skill-building

1. Find the domain and range of $y = 2 \sin^{-1}(x) + \pi$.
2. Sketch $y = \sin^{-1}(2x)$ and label the endpoints.
3. Determine the domain of $y = \cos^{-1}\left(\frac{x-1}{5}\right)$.
4. Given $f(x) = 2 \tan^{-1}(x)$, find the coordinates of the y -intercept and state the equations of the horizontal asymptotes.
5. A function is defined by $g(x) = \cos^{-1}(x) - \frac{\pi}{2}$. Find the x -intercept.
6. Simplify the expression $\sin(\sin^{-1}(0.4))$.
7. Solve $2 \cos^{-1} x = \pi$ for x .
8. Describe the transformations required to transform $y = \sin^{-1} x$ into $y = 3 \sin^{-1}(x - 2)$.
9. Find the domain of $y = \sqrt{\sin^{-1} x}$.
10. Sketch $y = -\cos^{-1}(x)$ and state its range.

Easier Exam Questions

- (Shore 2022 Q5) What is the domain of $y = 3 \cos^{-1}(2x + 1)$?
 - Domain $[-1, 0]$
 - Domain $(1, 0)$
 - Domain $[-\frac{1}{2}, 0]$
 - Domain $[\frac{1}{2}, 0]$
- (Riverview 2023 Q3) The domain for $f(x) = 3 \cos^{-1} 4x$ is:
 - $[\frac{-3}{4}, \frac{3}{4}]$
 - $[\frac{-1}{3}, \frac{1}{3}]$
 - $[\frac{-1}{4}, \frac{1}{4}]$
 - $[-1, 1]$
- (Hurlstone 2021 Q1) What is the domain of the function $y = 5 \cos^{-1}(4x)$?
 - $-\frac{5}{4} \leq x \leq \frac{5}{4}$
 - $-\frac{1}{5} \leq x \leq \frac{1}{5}$
 - $-\frac{1}{4} \leq x \leq \frac{1}{4}$
 - $-1 \leq x \leq 1$
- (Cheltenham Girls 2023 Q1) The graph of $y = g(x)$ is shown below.

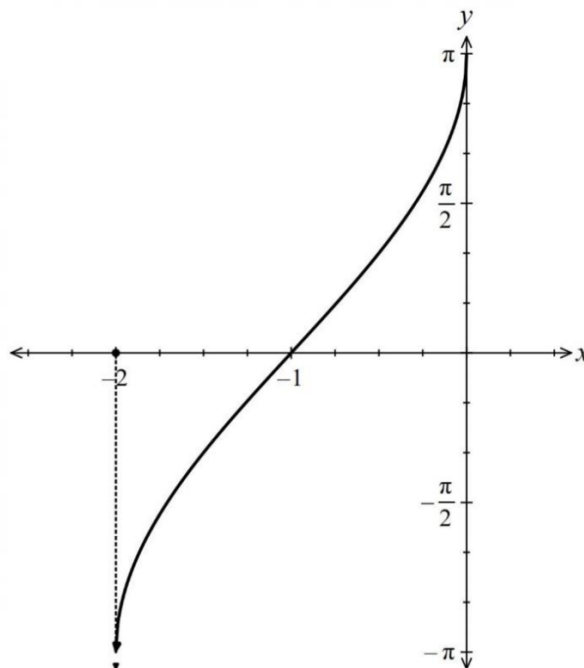


DIAGRAM NOT TO SCALE

Which equation could describe $g(x)$?

- $g(x) = \frac{1}{2} \cos^{-1}(x - 1)$

- (B) $g(x) = 2 \cos^{-1}(x + 1)$
- (C) $g(x) = \frac{1}{2} \sin^{-1}(x - 1)$
- (D) $g(x) = 2 \sin^{-1}(x + 1)$

5. (Manly 2023 Q1) What is the domain and range of $y = 4 \cos^{-1} \left(\frac{3x}{2} \right)$.

- (A) Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$ Range: $-4\pi \leq y \leq 4\pi$
- (B) Domain: $-\frac{3}{2} \leq x \leq \frac{3}{2}$ Range: $-4\pi \leq y \leq 4\pi$
- (C) Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$ Range: $0 \leq y \leq 4\pi$
- (D) Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$ Range: $0 \leq y \leq 4$

6. (Manly 2025 Q13a) The graph of $y = \sin^{-1}(x - 4)$ is dilated horizontally by a factor of 2 and then shifted right 1 unit.

Write the equation of the transformed graph.

2

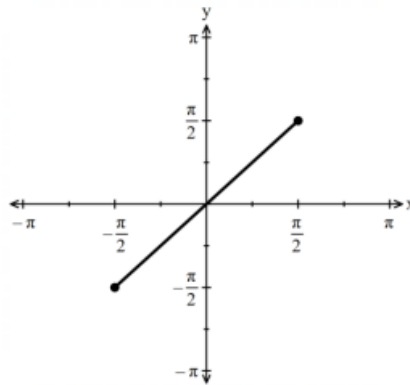
Harder Exam Questions

1. (Girraween 2024 Q15b) For the function $y = 2 \sin^{-1}(2x - 1) + \frac{\pi}{2}$

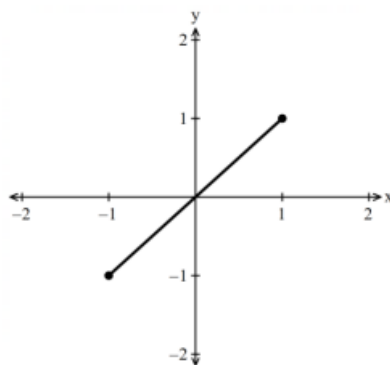
- (i) Find the domain and range. **2**
- (ii) Sketch the curve, clearly showing the coordinates of the endpoints. **2**

2. (Killara 2023 Q8) Which of the following sketches represents $f(x) = \cos^{-1}(\cos(x))$?

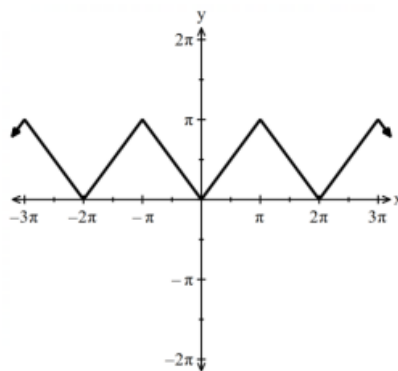
A.



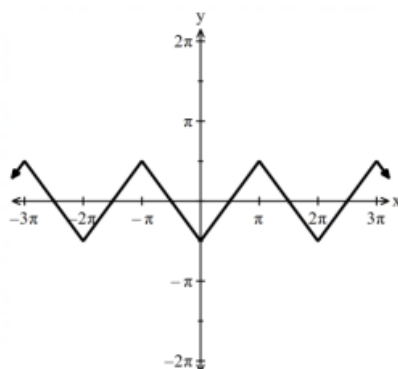
B.



C.



D.



3. (Knox 2021 Q11f) A function is defined by $f(x) = \sin^{-1}(1 - x)$.

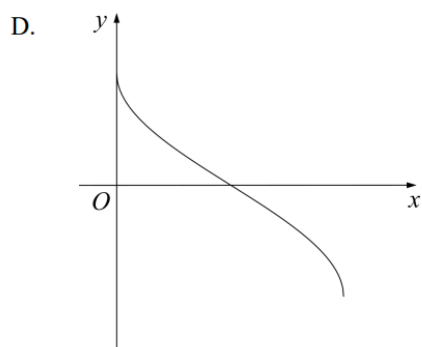
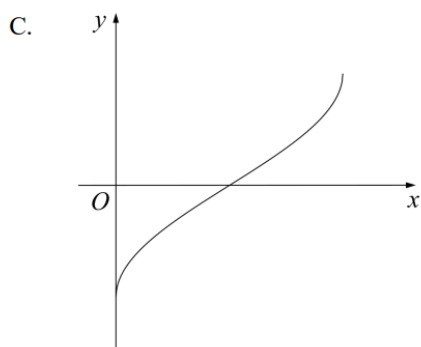
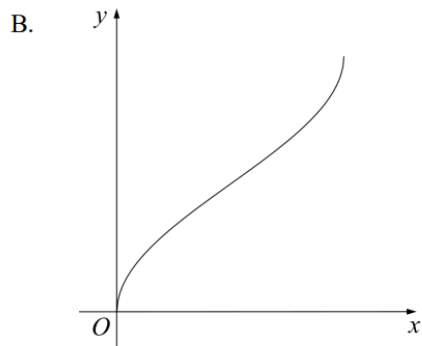
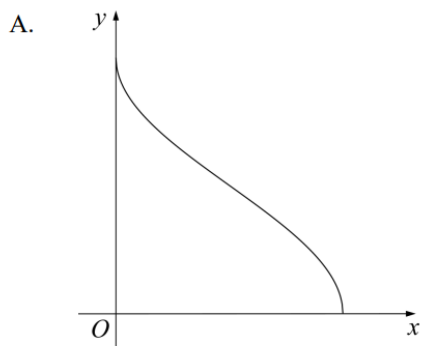
(i) Find the domain of $f(x)$.

1

(ii) Draw a neat sketch of $y = f(x)$.

1

4. (Abbotsleigh 2024 Q7) Which of the graphs below shows $y = 2 \sin^{-1} \left(1 - \frac{x}{3} \right)$?



5. (Caringbah 2020 Q12a)

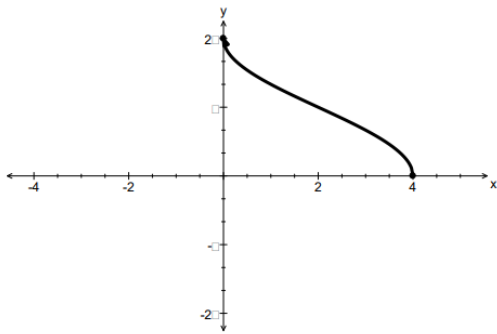
(i) Sketch the graph of the function $y = \tan^{-1} \frac{1}{2}(x - 2)$ showing clearly the intercepts on the axes and the equations of the asymptotes. **2**

(ii) The graph of the function $y = \tan^{-1} \frac{1}{2}(x - 2)$ is transformed by a translation left by 1 unit, then a horizontal dilation with a scale factor 2.

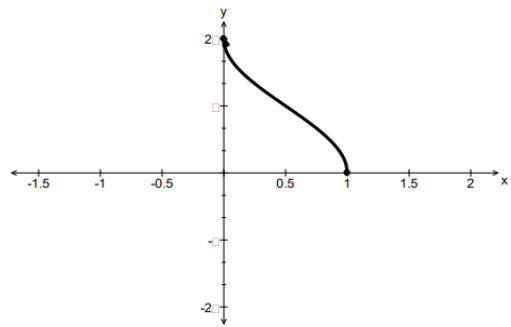
Find the equation of the transformed graph. **2**

6. (Newcastle Grammar 2020 Q8) Which of the graphs below shows $y = 2 \cos^{-1} \left(\frac{x}{2} - 1 \right)$?

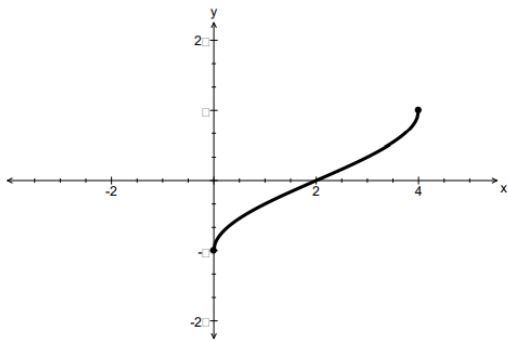
(A)



(B)



(C)



(D)

